

HDD & ROAD CROSSING GAS PIPELINE - DUAL-LAYER FBE (FBE+ARO) QUALITY PLAN (ITP)

Pipe & Coating Description: 30" (762 mm) x 12.70 mm, Grade X60M PSL 2 with Dual-Layer FBE Coating (Corrosion Base + Abrasion Overcoat ARO) | **Governing Codes:** CSA Z245.20 / NACE SP0394 / ISO 21809-2 / API Spec 5L 47th Ed.

No	Inspection / Test Stage	Frequency & Sampling	Location & Specimen	Test Standard & Procedure	Specified Acceptance Criteria & Tolerances	Governing Clause	Hold / Witness
1	FBE Powder & ARO Overcoat Verification	Each batch lot of Base FBE & ARO powder	Manufacturer quality certificates	CSA Z245.20 / ISO 21809-2	Base FBE: Gel time, moisture <= 0.50%, Delta Tg (-2 °C to +3 °C); ARO: High-abrasion/gouge resistant resin, EN 10204 3.1 cert	CSA Z245.20 Cl. 6	H/R
2	Bare Pipe Incoming & Blast Cleaning	100% of all pipes	External steel surface	ISO 8501-1 / SSPC-SP 10	Solvent clean (SSPC-SP 1) + Near-White Blast Sa 2.5; Sharp angular profile Rz 60 µm - 90 µm; Chlorides <= 20 mg/m²	CSA Z245.20 Cl. 8	H/W
3	Chemical Pre-Treatment & Induction Heating	Continuous monitoring all pipes	Full pipe circumference	Optical Pyrometer	Deionised water rinse + Chromate/Silane pre-treatment; Medium frequency induction heat: 230 °C - 245 °C (max 250 °C)	NACE SP0394 Cl. 4	H/W
4	Dual-Layer Electrostatic Spray Application	Continuous tandem spray booths	Outer pipe surface	Electrostatic powder guns	Booth 1: Base corrosion FBE (350 - 450 µm) -> Immediate Booth 2: Tough ARO overcoat (400 - 550 µm) before base gelation	CSA Z245.20 Cl. 9	W
5	Water Quench Cooling & Cure Assessment	100% pipes quench trough + DSC on sample	Pipe surface / cure sample	DSC Thermal Analysis (Tg)	Quench to < 90 °C; Thermal analysis DSC: Glass transition Delta Tg within -2 °C to +3 °C (confirming full cure)	CSA Z245.20 Cl. 12.7	H/W
6	100% High-Voltage Holiday Detection	Every coated pipe (100% full surface)	Full 360° coating exterior	NACE SP0188 / CSA Z245.20	High voltage spark tester: 5.0 V/µm of total thickness (approx 4.0 kV - 4.5 kV). ZERO holidays allowed per pipe	CSA Z245.20 Cl. 12.8	H/W
7	Total Coating Thickness (DFT) Verification	100% of all pipes (12 points around body)	Both ends, quarter points, middle	SSPC-PA 2 / CSA Z245.20	Total DFT: 750 µm - 1000 µm (Base FBE >= 350 µm, ARO Overcoat >= 400 µm). Minimum individual reading >= 700 µm	CSA Z245.20 Table 2	W
8	Cross-Section Porosity & Interface Rating	1 test per shift (on ring coupon)	Microscope cross section (30x)	CSA Z245.20 Cl. 12.10	Cross section foam porosity rating <= 3; Steel-to-coating interface porosity rating <= 2 (no continuous voids)	CSA Z245.20 Cl. 12.10	W
9	Cold Flexibility / Mandrel Bend Test	1 test per shift / lot	Laboratory bend strap at -30 °C	CSA Z245.20 Cl. 12.11	Bend deflection: 2.5° per pipe diameter at -30 °C; ZERO cracking, tearing, or disbonding on tension surface	CSA Z245.20 Table 4	H/W
10	Gouge Resistance & Impact Testing	Qualification / 1 test per 100 pipes	Laboratory coupon	ASTM G14 / CSA Z245.20	Impact resistance: >= 3.0 Joules at -30 °C (no holiday); Gouge depth: <= 0.25 mm under 50 kg hardened stylus load	CSA Z245.20 Cl. 12.12	H/W
11	Cathodic Disbondment Testing (28d & 24h)	1 test per lot / qualification batch	Laboratory electrochemical cell	CSA Z245.20 Cl. 12.9	28 days at 20 °C: Disbondment radius <= 6.5 mm; 24 hours accelerated at 95 °C: Disbondment radius <= 5.0 mm	CSA Z245.20 Table 4	H/W
12	Cutback, Stenciling & Final Acceptance	100% of all finished coated pipes	Pipe ends & exterior surface	Visual & Gauge check	Coating cutback: 75 mm +/- 15 mm; bevels clean and rust-inhibited; color coded green/blue band; EN 10204 3.1 cert	CSA Z245.20 Cl. 13	H/W